

TEKION INTERVIEW EXPERIENCE

About the Company:

Tekion, founded in 2016 by former Tesla CIO Jay Vijayan, is a fast-growing cloud technology company revolutionizing the automotive retail industry. Headquartered in Pleasanton, California with major tech hubs in India, Tekion serves over 2,000 dealerships and 52 OEM brands worldwide through its flagship Automotive Retail Cloud (ARC) platform.

Number of Rounds: 4

1. Online Assessment
2. Technical Interview - 1
3. Technical Interview - 2
4. HR Interview

Online Assessment:

MCQ - 20 Questions (DBMS, OS, OOPS, DSA)

DSA:

Problem 1:

You are given two arrays, marie and annie, which represent the face cards of two players, and a string flip that contains either "Even" or "Odd". The game is played in rounds, where each index i of the arrays represents one round.

- If flip = "Even", then the players calculate their scores as:
 - Marie's score for this round = $\text{abs}(\text{marie}[i] - \text{annie}[i])$
 - Annie's score for this round = $\text{abs}(\text{annie}[i] - \text{marie}[i])$
- If flip = "Odd", both players skip their cards for that round (no score is added).

After each round, the value of flip toggles (i.e., "Even" becomes "Odd", and "Odd" becomes "Even").

Your task is to find the final score of Marie and Annie after all rounds.

Example:**Input:**

marie = [4, 5, 6, 7]

annie = [1, 2, 3, 4]

flip = "Even"

Process:

- Round 1 (flip = "Even"):
Marie = $|4 - 1| = 3$
Annie = $|1 - 4| = 3$
- Round 2 (flip = "Odd"): skipped
- Round 3 (flip = "Even"):
Marie = $|6 - 3| = 3$
Annie = $|3 - 6| = 3$
- Round 4 (flip = "Odd"): skipped

Final Scores:

Marie = $3 + 3 = 6$

Annie = $3 + 3 = 6$

Problem 2: Bandwidth Allocation (0/1 Knapsack Variant)

You are given:

- An array `bandwidthNeeded[ ]`, where `bandwidthNeeded[i]` represents the amount of bandwidth required to process the *i*-th process.
- An array `requestsProcessed[ ]`, where `requestsProcessed[i]` represents the number of requests that can be processed if the *i*-th process is allocated its required bandwidth.
- An integer `totalBandwidth`, which represents the maximum total bandwidth available.

You need to allocate the available bandwidth to processes in such a way that:

- A process is either fully allocated its required bandwidth or not allocated at all (no partial allocation).
- The total allocated bandwidth does not exceed `totalBandwidth`.

- The total number of requests processed is maximized.

Your task is to determine the maximum number of requests that can be processed with the given total bandwidth.

Example

Input:

bandwidthNeeded = [2, 3, 4, 5]

requestsProcessed = [3, 4, 5, 6]

totalBandwidth = 5

Explanation:

- If we choose processes 1 and 2: bandwidth used = $2 + 3 = 5 \rightarrow$ requests processed = $3 + 4 = 7$
- If we choose processes 3: bandwidth used = $4 \rightarrow$ requests processed = 5
- If we choose process 4: bandwidth used = $5 \rightarrow$ requests processed = 6

The optimal allocation is processes 1 and 2 $\rightarrow$ 7 requests.

Output:

Maximum requests processed = 7

We took the online assessment on the HackerRank platform at CUIC. I managed to solve the first question completely and the second question partially, and I did well in the MCQ section. A total of 30 students from CSE were shortlisted for the next round.

Technical Interview 1 (50 minutes)

This round mainly focused on DSA and Core subjects.

- The interviewer first went through my resume and asked about my projects and the tech stacks I had worked with.
- Since I had mentioned NSS under campus involvements, he asked about that as well.
- He then asked me about my favorite subject. I answered DBMS, so he followed up with several DBMS-related questions.

DBMS Questions

1. What is replication, sharding, and partitioning?
2. What is the CAP theorem?
3. What is a transaction?
4. What are the ACID properties?
5. What are the different types of joins?
6. What is functional dependency?
7. What is a read/write conflict?
8. What is copy-on-write?

DSA Problems

He then moved on to DSA and asked me three problems:

1. Given an array and integer k, left shift the array by k times.
2. Find the parent node of a given node in a binary tree.
3. Find the intersection node of two linked lists.

I provided both brute-force and optimal approaches for all the problems and wrote the pseudo-code as requested. The interviewer was satisfied with my solutions.

In the middle of the interview, I was also asked to introduce myself.

OS Questions

Later, the interviewer asked if I was familiar with OS concepts. Since I said yes, he asked the following questions:

1. What are semaphores?
2. What is a mutex?
3. What is a process?
4. What is a thread?
5. What are paging and segmentation?
6. What is a page fault?
7. What is thrashing?

8. What is a deadlock?

Finally, the interviewer asked whether I had any questions. I asked two questions in return.

Technical Interview 2 (45 minutes)

For the first 10 minutes, the interviewer introduced himself and spoke about the company. After that, I was asked to introduce myself.

This round was mainly focused on projects.

- He asked about my favorite project, and I explained my CIP project.
- He asked several follow-up questions, such as:
 - What tech stack did I use?
 - Why did I choose that particular stack?
 - Have I worked with AWS, and what services do I know?
- He also gave me a few scenarios and asked how my application would behave in those cases.
- I was then asked to design an algorithm for a scenario related to my project and explain its time complexity.

The interviewer also asked a DSA problem:

Problem:

Given an array of strings, remove all the strings that have a permutation of themselves present in the array.

I provided both a brute-force and an optimal approach and discussed the time complexity for each.

Towards the end, he asked a few questions from Operating Systems (OS) as well.

Finally, the interviewer asked whether I had any questions, and I asked a couple of questions in return.

HR Interview (15 minutes)

This round was more of a casual conversation.

- The interviewer first introduced herself.

- She then asked me to introduce myself.
- I was asked whether I had attended the pre-placement talk.
- She asked what I think about the company and what makes me a good fit for the role.
- I was also asked about the key takeaways from my projects and to explain all of them briefly.
- She asked me to share any conflicts I had faced with my teammates and how I managed them.
- Finally, she asked whether I had any questions, and I asked a couple in return.

At the end of the process, 9 students from CSE were selected, and I was one of them.

Key Notes:

- Revise your core subjects like DBMS, OS, Computer Networks, and DSA thoroughly before the interview.
- Practice solving coding problems and be prepared to explain both brute force and optimal approaches.
- Always mention the time and space complexity when explaining your solutions.
- Be confident in your projects and clearly explain the technologies used, design choices, and outcomes.
- Communicate your answers in a structured and confident manner.
- Prepare for behavioral questions such as teamwork, conflict management, strengths, and weaknesses.
- Stay calm during the interview and admit honestly if you don't know something, while trying to reason it out.
- Show genuine interest in the company by asking thoughtful questions at the end of the interview.

Feel free to reach out:

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